

Nicolas Espinosa Dice

Education

2023– Present **Ph.D., Computer Science, Cornell University**
Advisor: Wen Sun

2022 **B.S., Mathematics and Computer Science, Harvey Mudd College**
Graduated with High Distinction and Departmental Honors in Mathematics and Computer Science.

Research Experience

2023– Present **Department of Computer Science, Cornell University**
Advisor: Wen Sun
Currently researching imitation learning and reinforcement with generative models.

2020–2022 **AMISTAD Lab, Harvey Mudd College**
Advisor: George Montanez
Developed a probabilistic model of abductive reasoning, and derived generalization error bounds of learning algorithms in an algorithmic search framework.

Industry Experience

2022–2023 **Research Scientist, Dasion**
Advisor: Weiqing Gu
Developed an audio-based model capable of detecting Autism Spectrum Disorder in children, contributing to a successful NSF SBIR Phase I Project Pitch, and co-authored successful NSF SBIR Phase II Proposal.

2021–2022 **Project Lead, Clinic Program, Harvey Mudd College**
Project lead of undergraduate team working with Rockerbox that built an anomaly detection model on time-series marketing data.

Summer 2021 **Software Engineer Intern, Etsy**
Developed a transformer-based deep learning model with DistilBERT architecture using Tensorflow to classify safe search queries.

Summer 2019 **Software Engineer Intern, Viasat**
Built a heads-up display on a Microsoft HoloLens, integrated with data from Link 16 radio network. Presented by Viasat at AUSA 2019.

Publications

Nicolas Espinosa Dice, Megan Kaye, Hana Ahmed, and George Montañez. A probabilistic theory of abductive reasoning. In *Proceedings of the 13th International Conference on Agents and Artificial Intelligence*, volume 2, 2021.

Ramya Ramalingam, **Nicolas Espinosa Dice**, Megan L Kaye, and George D Montañez. Bounding generalization error through bias and capacity. In *2022 International Joint Conference on Neural Networks (IJCNN)*, pages 1–8. IEEE, 2022.

Nicolas Espinosa Dice, Sanjiban Choudhury, Wen Sun, and Gokul Swamy. Efficient imitation under misspecification. *Preprint*, 2024.

Nicolas Espinosa Dice, Gokul Swamy, Sanjiban Choudhury, and Wen Sun.

Efficient inverse reinforcement learning without compounding errors. In *First Reinforcement Learning Safety Workshop*, 2024.

Teaching and Outreach

- 2024 **Teaching Assistant**, *Foundations of Reinforcement Learning*, Cornell University
- 2024 **ASCEND Mentor**, *Department of Computer Science*, Cornell University
- 2024 **Graduate Student Mentor**, *Department of Computer Science*, Cornell University
- 2022–2024 **College Mentor**, *UStrive*
- 2018–2022 **Chair**, *Honor Board*, Harvey Mudd College
- 2018–2022 **Public Outreach Director**, *Society of Professional Latinx in STEM*, Harvey Mudd College
- 2021 **Grader and Tutor**, *Artificial Intelligence*, Harvey Mudd College

Fellowships and Awards

- 2023 **Hopper-Dean and Bowers Fellowship**, *Cornell University*
- 2023 **Dean's Scholar**, *Cornell University*
- 2022 **Departmental Honors in Computer Science**, *Harvey Mudd College*
- 2022 **Departmental Honors in Mathematics**, *Harvey Mudd College*
- 2022 **Don Chamberlain Computer Science Research Award**, *Harvey Mudd College*
Second student at Harvey Mudd to win both Chamberlain Research and Clinic Program Individual Awards
- 2018 **Harvey S. Mudd Merit Award**, *Harvey Mudd College*